

14 — Section Grammar (Home Layout)

Status: DRAFT v0.1 (2026-06-12)

Describes how a role's home page is laid out. This is the only file about layout.

It is also common to all roles: same grammar, different sections.

Purpose

The persona card (10) says *what* a role cares about (KPIs, insights, questions, actions). It does not say how the home page is arranged.

This file defines a **home template**: an ordered list of sections.

Each section pulls from the card and from the resolved data.

We keep layout separate from the card on purpose (locked decision). The card is the meaning. The home template is the arrangement. Both are common across roles.

Note on visuals

This grammar is about **business structure**, not pixels. No CSS, colour, or widths.

The frontend decides the look. This file only says "show a summary here, insights next".

Design rules

- A home template targets one `card_id` .
- Sections are ordered. The order is the page order.
- A section either lists explicit metrics, or pulls from the card / engine output.
- It never invents data. It only arranges what 10 , 12 , and 13 produce.

Section types

type	shows	pulls from
summary	a few headline numbers	explicit metric ids
kpi_strip	a row of KPIs	explicit metric ids (or card.kpis)
insight_list	ranked insight cards	card.insight_types + engine ranking
action_list	recommended actions	the card's actions whose from_insight matches the top insights (or explicit)

type	shows	pulls from
suggested_questions	Ask-Analyst questions	card.questions (question bank)
chart	one trend chart	a metric (+ optional trend_grain ; defaults to the sales_trend preference)

Field table

Field	Type	Required	Notes
card_id	string	yes	which persona this home is for; must exist in 10
sections	Section[]	yes	ordered list
sections[].id	string	yes	stable id
sections[].title	string	yes	shown as the section heading
sections[].type	enum (above)	yes	which section kind
sections[].items	Item[]	summary/kpi_strip	{label, metric} ; format comes from 12
sections[].max_items	integer	list types	cap on rows
sections[].filters	object	insight_list	e.g. {priority: [critical, high]}
sections[].insight_types	string[]	no	override; default = card.insight_types
sections[].source	enum priority_insights question_bank	action_list / suggested_questions	where to pull
sections[].metric	string	chart	metric id for the chart
sections[].trend_grain	enum day week month	chart (optional)	overrides the resolved sales_trend preference (11)

JSON Schema

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://cpg-analytics/spec/section-grammar.schema.json",
  "title": "HomeTemplate",
  "type": "object",
  "additionalProperties": false,
  "required": ["card_id", "sections"],
  "properties": {
    "card_id": { "type": "string", "pattern": "^[a-z][a-z0-9_]*$" },
    "sections": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "object",
        "additionalProperties": false,
        "required": ["id", "title", "type"],
        "properties": {
          "id": { "type": "string" },
          "title": { "type": "string", "minLength": 1 },
          "type": { "enum": ["summary", "kpi_strip", "insight_list",
                           "action_list", "suggested_questions", "chart"] },
          "items": {
            "type": "array",
            "items": {
              "type": "object",
              "additionalProperties": false,
              "required": ["label", "metric"],
              "properties": {
                "label": { "type": "string" },
                "metric": { "type": "string" }
              }
            }
          }
        }
      },
      "max_items": { "type": "integer", "minimum": 1 },
      "filters": { "type": "object" },
      "insight_types": { "type": "array", "items": { "type": "string" } },
      "source": { "enum": ["priority_insights", "question_bank"] },
      "metric": { "type": "string" },
      "trend_grain": { "enum": ["day", "week", "month"] }
    }
  }
}
```

Validation rules (check in CI)

1. `card_id` exists in the persona card set (10).
2. Every `items[].metric` and any `chart metric` exists in the metric registry (12).
3. Any `insight_types` listed exist in the insight-rule registry (13) and are a subset of the card's `insight_types` .
4. `summary / kpi_strip` sections must have `items` . List sections should have `max_items` .

Worked example

See `examples/nsm.home.yaml` . It rebuilds the National Sales Manager home you sketched:
Today's Business Summary, Top Priority Insights, Recommended Actions, Ask CPG Analyst.